

You did not build a classifier. You built an **LLM-as-a-Judge** — and half of its reliability layer is already on disk.

Applying **Gu et al., “A Survey on LLM-as-a-Judge”** (arXiv:2411.15594v6, Oct 2025) to the veterinary radiology label pipeline — then re-scoring the Vetology pitch in light of what the survey says is missing.

ALREADY BUILT

3 / 10

components of the survey’s reliability wrapper `f_R` — plus a fourth, partial

FREE TO ADD

7 / 13

paper→code mappings cost nothing — the data is already on disk

PAID PROBES

<\$2

bias, robustness and ablation runs across 50 cases

THE OPENING

\$4.1

the survey names your exact study as an under-researched gap

The survey’s formalism, with your filenames in it

Two definitions. You have implemented the first one completely. The second — the reliability wrapper — is the one worth building.

BASIC DEFINITION · §2

$$E \leftarrow P_{LLM} (X \oplus C)$$

Fully implemented. Every symbol maps onto a file you have already written.

X the free-text radiologist report —
`RadiologyCase.findings_text + .conclusions_text`

C `prompts/{category}.json` + the injected finding list + the schema’s `Field(description=...)` strings

P `llm.build_model()` — openai · kimi · groq · mistral

E `19 / 19 / 10 {finding, label, evidence, reasoning} per category, enum-constrained`

RELIABILITY DEFINITION · §2

$$R \leftarrow f_R (P, X, C)$$

“constraints and validation methods applied systematically ... to mitigate biases, control variability, and confirm robustness against adversarial inputs.” — this is the layer you have no name for, and it is half built.

COMPONENT OF F_R	STATUS
Constrained / structured output §2.3.1	BUILT
Variability control — N=5 replicates §3.3.1	BUILT
Confidence signal — CE §2.1.1	BUILT
Agreement with human judgment §4.1	PARTIAL
Multi-model integration §3.3.1	\$0 · MISSING

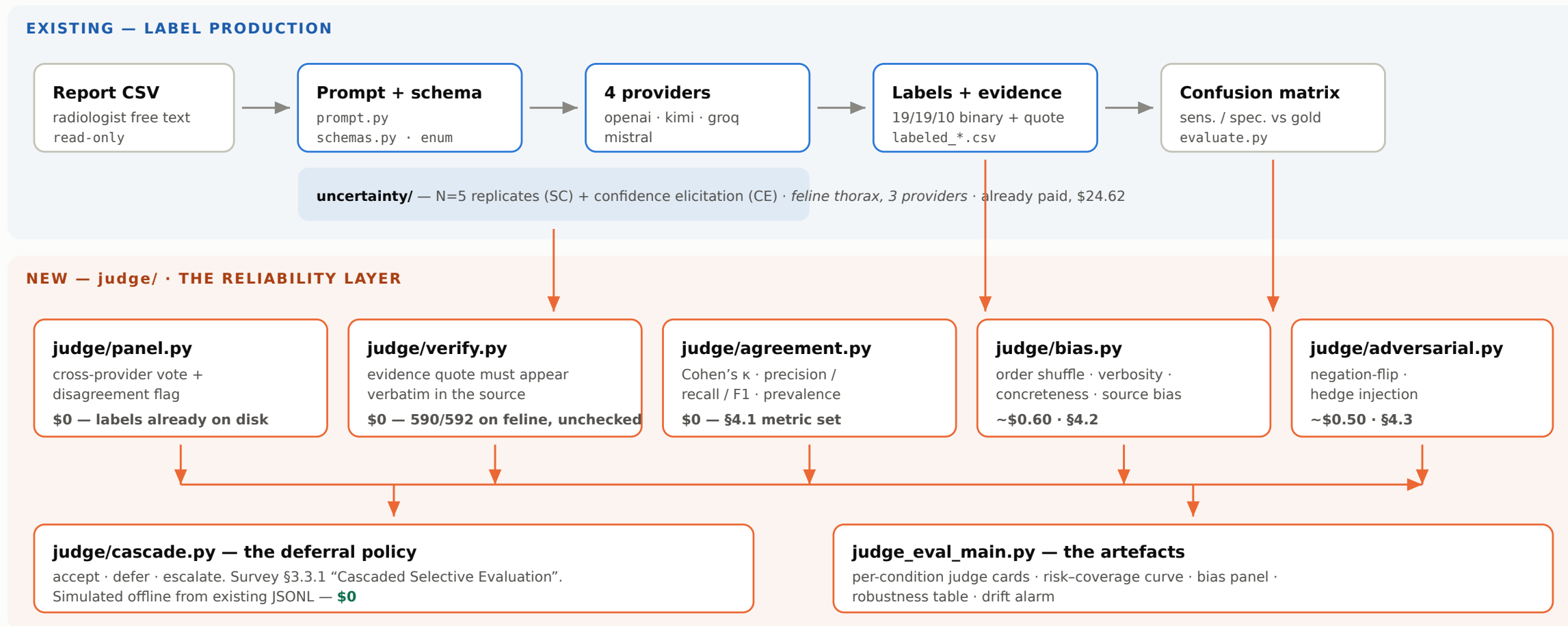
COMPONENT OF F_R	STATUS
Self-verification of evidence §3.3.2	\$0 · MISSING
Cascade / selective evaluation §3.3.1	\$0 · MISSING
Drift & temporal consistency §6.5	\$0 · MISSING
Bias measurement — 12 named types §4.2	~\$0.60
Adversarial robustness §4.3	~\$0.50

And the survey names your exact study as an open gap — §4.1

*“Current meta-evaluation primarily focuses on LLM-as-a-judge **for models**, while there is a lack of sufficient meta-evaluation when these LLM evaluators are used for automatically annotating large-scale datasets. We advocate for more rigorous assessment of the alignment between LLM-as-a-judge and human judgment when they are employed for large-scale data annotation.” — you are on the **for data** branch. Every table you produce from here aims at a stated gap rather than a crowded benchmark.*

Today's pipeline, and where the judge layer bolts on

A new `judge/` package, sibling to `classifier_multi/` and `uncertainty/`. Same conventions: pure functions, injected model, and the paid stage separated from the iterated stage by a file.



Paper → codebase, priced

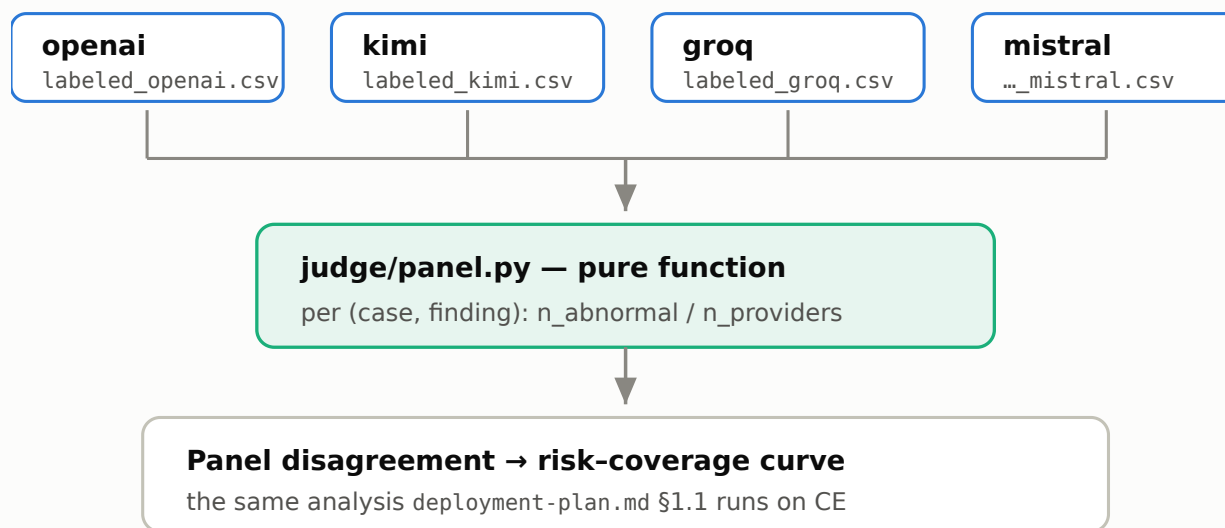
Eleven of the thirteen mappings, ranked by cost. Seven cost nothing because the data already exists. Full detail and file-level notes in `llm-as-a-judge-plan.md` §2.

SURVEY	WHAT IT PRESCRIBES	WHERE YOU STAND	BUILD	COST
§4.1	Kappa, % agreement, precision/recall	sensitivity/specificity only — and all-normal already scores 95.1%	<code>judge/agreement.py</code>	\$0
§3.3.1	Integrate multiple LLM evaluators by vote	10 labelled CSVs from 4 providers, never compared to each other	<code>judge/panel.py</code>	\$0
§3.3.2	Self-verification — keep only what passes	evidence quotes collected, never checked	<code>judge/verify.py</code>	\$0
§3.3.1	Cascaded Selective Evaluation (Jung et al.)	designed in <code>deployment-plan.md</code> §1.3, unbuilt — now with a citation	<code>judge/cascade.py</code>	\$0
§4.2.2	Length & concreteness bias	testable offline: report length / numeric density vs abnormal-call rate	<code>judge/bias.py</code> offline half	\$0
§6.5	Evaluation drift across model versions	<code>-latest</code> unpinned; no <code>model_id</code> in output rows	stamp id + freeze regression set	\$0
§2.3.2	Logit normalisation / score smoothing	you measured all three providers refuse logprobs — a result, not a gap	report it	DONE
§4.2.2	Position bias — shuffle and re-score	findings always sent in fixed column order; never tested	shuffle <code>finding_list</code> , 50 cases	\$0.30
§4.2.1	Self-enhancement / source bias	your CSVs hold the AI report <i>and</i> the radiologist report per case	label both, compare	\$0.30
§4.3	Adversarial robustness	clinical analogue already identified: hedges and negation scopes	<code>judge/adversarial.py</code>	\$0.50
§3.1.1	Decomposition of evaluation criteria	<i>prediction</i> : colitis/gastritis FPs are <i>definition</i> errors, not reading errors — the ablation tests it	one-line definition per condition + ablate	\$0.50

The panel: a new result, tomorrow, for \$0

Ten labelled CSVs on identical cases — four providers on feline thorax, three each on canine thorax and canine abdomen. The survey calls combining them *integrating multi-source evaluation results* and cites three systems that do it by vote. Nobody has done it here.

ON DISK TODAY — ALREADY PAID FOR



THREE OUTCOMES — NO LOSING BRANCH

Beats CE → a free proxy outperforms the one that cost \$24.62.

Ties CE → independent confirmation; cheaper CE wins on cost.

Loses → provider diversity adds nothing here. Don't buy four providers.

Whichever way it falls, it answers a question you have not asked yet — for nothing.

Why this answers something SC could not

Sample consistency was high-precision, low-coverage. `catalog.md`
D14: 98% of rows were unanimous across five replicates, so the detector only fires on ~2% of rows.

That is *within-model* variation, and reasoning models doing closed-set extraction have almost none of it. **Cross-model disagreement is a different signal** — four models with different training data fail in different places.

UNANIMOUS ACROSS 5 RUNS

98%

so SC fires on 2 rows in 100

ACCURACY WHEN SPLIT

0.39-0.74

across the three providers, vs ~0.98 when unanimous — a sharp, rare signal

Also free, same day

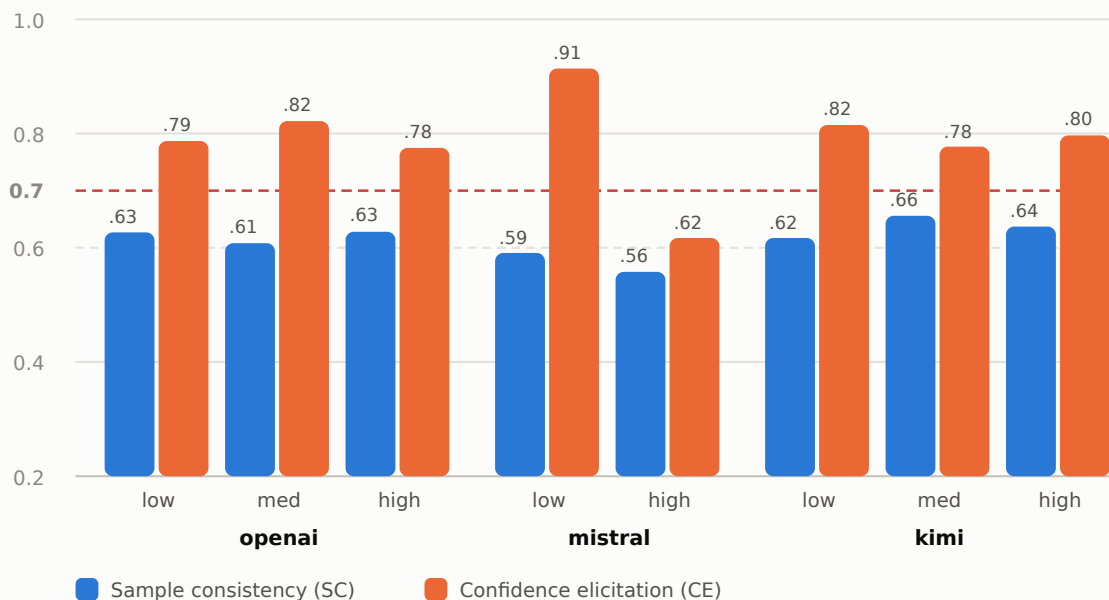
- Cohen's κ per condition, with prevalence beside it
- Evidence-grounding check across all nine `reasoning_*.json`
- Length & concreteness bias correlations
- Cascade cost simulation from existing JSONL

Three results you already own, re-anchored to the survey

Costs nothing. Turns three findings that currently read as “what happened” into three findings with a mechanism and a citation.

Confidence elicitation beat sample consistency in every cell

ROC AUC by provider and effort tier · `catalog.md` D14 · higher is better · the dashed line is 0.70, the usefulness threshold Savage et al. use



Read the caveat with the chart. Accuracy is ~97%, so each AUC rests on roughly 25–30 errors per cell, and the CIs say so (mistral/high CE: 0.511–0.717). SC clears 0.70 in **zero of eight** cells; CE clears it in **seven of eight** — only mistral/high misses. The ordering is robust; the individual heights are not.

D14 R1 • CE beats SC — reversing Savage et al.

The survey says absolute self-scoring is *less* robust than relative comparison (§3.1.1), which predicts CE should be the weak proxy. It was the strong one.

Mechanism: closed-set extraction removes the answer variability SC needs to measure. A result that contradicts two sources *and* has an explanation is worth more than one that agrees with both.

D14 R2 • SC is a detector, not a ranker

AUC rewards ranking across the whole distribution; SC is constant on 98% of rows. **Report precision/recall at the “not unanimous” threshold instead of AUC.** Your own `catalog.md` open item says this — §3.3.1 vs §4.1 explains why it happens.

D5 • No provider exposes token logprobs

The survey treats logit normalisation and score smoothing (§2.3.2, §3.3.2) as standard post-processing, noting they need token-probability access. You measured that three 2026-era reasoning models no longer provide it.

Constrained decoding and logit access are in tension — and the vendors chose constrained decoding.

What they do, and where the pressure is

CLASSIFIERS

91

27 canine thorax · 25 canine abdomen · 15 feline thorax · 18 feline abdomen · 6 spine/MSK

VALIDATION BASE

300k

multi-image patient cases from real practices, “not synthetic datasets”

PUBLISHED METRICS

4 → 11

per classifier, in ten weeks (Jan → Mar 2026)

RETRAINED IN ONE RELEASE

31

models, 31 Mar 2026 — plus new classifiers monthly

GROUND TRUTH

DACVR

US board-certified veterinary radiologists

“Complete transparency isn’t a competitive advantage we’re protecting, it’s a professional obligation.” — Eric Goldman, President, on publishing full classifier performance metrics, January 2026

THREE READINGS THAT MATTER

1 · The bottleneck recurs, forever

31 retrains in one month; new classifiers monthly. Every retrain needs a freshly scored test set, and every score is a human turning a free-text radiologist report into per-condition labels. **That cost is re-paid every release cycle** — and it is the rate limiter on how fast they ship.

2 · They keep raising their own bar

4 → 11 metrics in ten weeks is a company that will keep adding metrics. All eleven measure *the classifier*. **None measures the reliability of the labelling that produced the ground truth they are scored against.**

3 · An LLM already ships into their clinic

The radiologist platform offers “a custom language model” trained on a practice’s historical cases that drafts conclusions and recommendations. **Nothing public describes a confidence, grounding, or deferral layer on it.**

Other context worth carrying into the call

Integrations — ezyVet, DaySmart Vet, VetRocket; AI scribes ScribbleVet (Oct 2025) and CoVet (Jan 2026) push SOAP notes straight into radiology requests. **Validation partners** — AMC New York and Tufts University, with a public invitation for independent validation. **Three products** — AI screening, a white-labelled radiologist platform, and board-certified read services.

State this as inference, not fact

The bottleneck reading is a **strong inference** from their public performance page, their release cadence, and the shape of your data — not something they have confirmed. Ask on the call before building a proposal on it.

Four ways to add value, ordered by public commitment

IDEA 1 · HIGHEST CONFIDENCE

Be the measurement instrument behind the transparency programme

Their problem. A permanent, growing annotation cost that gates classifier throughput.

You bring. A judge that reproduces the labelling step *plus* a measured deferral policy — one extra CE call per finding, defer below 0.90: **62% auto-accepted catching 85% of all errors** (Kimi) or **91% auto-accepted catching 46%** (OpenAI). Same mechanism, two operating points — always quote the catch rate beside the auto rate.

Why it lands. It doesn't ask them to trust your model over theirs — it asks them to trust it over their annotation backlog.

The ask. Stratified enrichment: $n \geq 20$ positives per condition. A query against their case store, not a modelling change — and only they can authorise it.

IDEA 3 · ATTACHES TO A SHIPPING FEATURE

A confidence + grounding layer on their radiologist-platform LLM

Their problem. A generated conclusion arrives with no signal about which sentences are grounded in the findings and which the model inferred — so the radiologist must read every word with equal suspicion, erasing the time saving.

You bring. Evidence grounding (§3.3.2): every claim quotes its source, checked verbatim — **you already collect these quotes and never test them**. Plus cascaded selective evaluation (§3.3.1) for bounded cost and latency.

Why it lands. It argues for *more* radiologist authority, which is what their own copy says they care about.

IDEA 2 · ON-BRAND

Metric #12 — reliability of the ground truth itself

Their problem. Every sensitivity figure inherits an unmeasured error bar from a single, unadjudicated annotation pass.

You bring. The survey's §4.1 metric set applied to the *annotation layer*: Cohen's κ between judge and human, per condition, with prevalence — plus the §4.2 bias audit and §4.3 robustness suite. Nobody in veterinary AI publishes either.

Caveat to state first. Agreement is not truth. κ measures agreement between two raters; it does not make either right. Saying that unprompted is what stops the idea being over-sold.

IDEA 4 · MENTION, DON'T BUILD

Disagreement triage — turn evaluation exhaust into a work queue

AI report and radiologist report exist for the same case. Label both, diff them, rank by panel confidence. Three products from one pipeline: a **QA queue** (confident disagreements, ranked), a **drift monitor** (run the frozen set after each monthly release, alarm on a κ shift — the survey's §6.5 *evaluation drift*), and a **training signal** (adjudicated corrections grow the gold standard, so the system gets cheaper the longer it runs).

Show the diagram. Do not promise the product in ten days.

Presentable → scalable → deployable

A · MAKE IT PRESENTABLE

The judge card

One row per condition — prevalence, TP/FN/TN/FP, sensitivity + CI, κ , panel agreement, deferral threshold, `model_id`, `prompt_version`, last run. **Laid out in the same column order Vetology publishes theirs**, so it reads as familiar in four seconds.

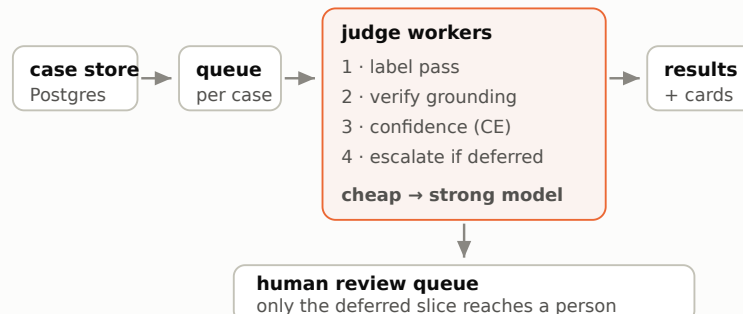
The single-case view — the thing they click

Paste a report → the source text comes back with **every evidence quote highlighted in place**, colour-coded by label, with a confidence bar and review flag per condition. Nearly free to build: the schema already returns a verbatim quote, so highlighting is `str.find`. It answers the survey's §6.4 complaint that judges are opaque black boxes.

Retire “accuracy” from every headline

All-normal scores **95.1%** on your data. Lead with prevalence, then κ , then precision/recall.

B · SCALE IT



Five levers, in order of effect

- **Cascade before you scale.** 91% auto-accepted at two calls. Sizing a fleet before applying the cascade sizes for work you’re about to delete.
- **Batch API** — ~50% off for 24h-tolerant work; a nightly revalidation job is exactly that shape.
- **Prompt caching** — cached input is 10x cheaper and the system prompt is identical across every case in a category.
- **Category as data, not code** — `categories.py` already makes a new study type one entry. 5 → 91 becomes data entry.
- **Idempotency you already have** — append-only JSONL with shortfall counting generalises straight to a queue key.

C · DEPLOY IT

There is no model artefact to deploy

No weights, no `model.pkl`. “Deploy the model” means **stand up a service and control cost and failure**. Saying that plainly is itself a signal.

GAP	EFFORT
FastAPI <code>/v1/label</code> , <code>/v1/judge</code> , <code>/v1/health</code>	1 day
Dockerfile	2 h
GitHub Actions over the existing 156 tests	1 h
<code>model_id</code> + <code>prompt_version</code> in every row	30 min
Structured JSON logs — case, tokens, cost, latency	3 h
Postgres instead of JSONL (only past the demo)	1 day

Do not build

Kubernetes · a vector DB · model serving infra · a fine-tuning pipeline. None is justified by this workload, and reaching for them signals the opposite of judgement.

Ten working days, under \$8 — and what would sink it

DAY	WORK	OUTPUT
1	<code>judge/agreement.py</code> + <code>judge/panel.py</code> — κ and the cross-provider panel over existing CSVs	the free result: panel vs CE risk-coverage
2	<code>judge/verify.py</code> over all nine <code>reasoning_*.json</code> ; per-condition error taxonomy	grounding pass rate; the colitis/gastritis definition-error claim, proven or dropped
3	Offline bias half (length, concreteness); freeze the regression set; stamp <code>model_id</code>	bias correlations; drift harness
4	Paid probes — order shuffle (\$0.30), source bias (\$0.30), negation-flip + hedge injection (\$0.50)	~\$1.10; the bias + robustness tables
5-6	FastAPI, Dockerfile, GitHub Actions over the existing 156 tests	callable service, green CI badge
7	Single-case demo UI with evidence highlighting; synthetic reports; free-tier deploy	the public URL they click
8	Judge cards + risk-coverage page, generated not hand-assembled	the artefact that mirrors their performance page
9	Two-page proposal — their bottleneck, your instrument, the stratified-sampling ask	the document that gets the second conversation
10	README, architecture diagram, buffer	a repo a stranger can read

Budget: order-shuffle \$0.30 · source-bias \$0.30 · adversarial \$0.50 · criteria ablation \$0.50 · synthetic reports \$1 · demo traffic \$5 · hosting \$0 — **under \$8**. Demo runs on synthetic reports throughout; every real number stays in a private appendix.

Limitations to state before anyone asks

- **50 cases, one species, one modality.** These are demonstrated methods with measured effect sizes — not validated thresholds. Don't let a chart imply otherwise.
- **κ is hostile to low prevalence.** At 5% base rates it is unstable and can look poor where accuracy looks excellent. Report n and prevalence beside every κ .
- **Agreement is not truth.** The judge reads another radiologist's finished text and cannot see the image. FP=26 vs FN=1 is the signature of a model *inferring* rather than reporting — which may be exactly what the persona asked for.
- **The Vetology bottleneck is inferred**, strongly, from their public pages and your data's shape. Ask on the call before building a proposal on it.
- **The survey is a survey.** It supplies vocabulary, taxonomy and citations — not a method that will lift your F1. The honest frame is “this is the thing nobody measures”, never “the paper says this will work”.

Two traps in the repo

The prompt lives in two places — the JSON file *and* the `Field(description=...)` strings, which ship inside the schema. An ablation that edits only the JSON has not fully changed the prompt.

Tuning the deferral threshold on the same 50 cases you measure it on makes the number optimistic. At $n=50$ a held-out split is barely viable — say so rather than quietly skipping it.